

Mingyun Kang

Researcher in Construction Management & Automation

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Education

Seoul National University of Science and Technology

Seoul, Republic of Korea

M.S. in Architectural Engineering (Integrated Bachelor-Master Program)

Mar 2024 – Aug 2025

- GPA : 4.22 / 4.5
- Thesis : Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System (Advisor : Prof. Taehoon Kim)

Seoul National University of Science and Technology

Seoul, Republic of Korea

B.S. in Architectural Engineering, School of Architecture

Mar 2017 – Feb 2024

- Double Major: Business Administration
- GPA : 3.8 / 4.5

Research Experience

Post-Master's Researcher

Goyang, Republic of Korea

Korea Institute of Civil Engineering and Building Technology (KICT)

Nov 2025 – present

Post-Construction Evaluation Center — Systematically evaluating outcomes of publicly funded SOC

- Compliance & Implementation Support: Developing research frameworks to encourage clients to fulfill post-construction evaluation requirements and to facilitate smooth reporting workflows.
- Feedback Loop & Knowledge Management: Designing mechanisms to extract structured insights from collected evaluation data, enabling lessons learned from completed projects to feed back into future initiatives and build a virtuous cycle of construction knowledge.

Research Assistant

Seoul, Republic of Korea

CONICT Lab, Seoul National University of Science and Technology

Dec 2021 – Aug 2025

Conducted research on computer vision-based automated construction supervision, 3D reconstruction, and human-robot collaboration at construction sites.

- Published 3 SCIE journal papers and 2 SCOPUS conference papers as first/co-author
- Filed 5 domestic patent applications related to construction automation and vision-based inspection
- Participated in 3 national R&D projects funded by NRF and the Ministry of Land, Infrastructure and Transport (MOLIT)

R&D Projects

Development of Human-Robot Collaboration Technology and Integrated Operation System for High-Altitude Work at Construction Sites

Apr 2025 – Dec 2027

Ongoing national R&D project (KRW 17,000M) funded by MOLIT developing XR-based human-robot collaboration platforms for multipurpose high-altitude construction work.

Digital Mapping-based Platform Research Laboratory for Healthy Construction Lifecycle

June 2022 – Feb 2025

National R&D project exploring digital mapping-based platforms for the full construction lifecycle. Focused on computer vision applications for automated construction supervision.

Development of Digital-based Architectural Supervision and Construction Automation Robot Technology

Apr 2022 – Dec 2026

Large-scale national R&D project (KRW 24,490M) funded by MOLIT aimed at developing digital architectural supervision systems and construction automation robots.

Publications (SCIE)

- Computer Vision-Based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System** Dec 2024
Applied Sciences, 15(1), 125.
Mingyun Kang, Sebeen Yoon, Taehoon Kim
[10.3390/app15010125](https://doi.org/10.3390/app15010125)  (Applied Sciences)
- Approach to Enhancing Panoramic Segmentation in Indoor Construction Sites Based on a Perspective Image Segmentation Foundation Model** Apr 2025
Applied Sciences, 15(9), 4875.
Juho Han, Sebeen Yoon, **Mingyun Kang**, Taehoon Kim
[10.3390/app15094875](https://doi.org/10.3390/app15094875)  (Applied Sciences)
- Automated Vision-based Location Monitoring for Jack Support** Jan 2025
KSCE Journal of Civil Engineering, 100374.
Sebeen Yoon, **Mingyun Kang**, Minhyung Kim, Taehoon Kim
[10.1016/j.kscej.2025.100374](https://doi.org/10.1016/j.kscej.2025.100374)  (KSCE Journal of Civil Engineering)

Publications (International Conference)

- Comparative Study of Structure from Motion on Construction Site** July 2025
ISARC Vol. 42, pp. 1395-1400. IAARC Publications. (SCOPUS)
Mingyun Kang, Sangmin Lee, Sebeen Yoon, Taehoon Kim
[10.22260/ISARC2025/0180](https://doi.org/10.22260/ISARC2025/0180)  (ISARC 2025 — International Symposium on Automation and Robotics in Construction)
- Approach for Improving Panoramic Image Segmentation Quality on Construction Sites using SAM** June 2024
CCC 2024, Prague, Czech Republic.
Juho Han, Sanghyeon Na, **Mingyun Kang**, Sebeen Yoon, Taehoon Kim
(Creative Construction Conference (CCC 2024))
- Instance Segmentation of Exterior Insulation Finishing System using Synthetic Datasets** June 2024
ISARC Vol. 41, pp. 1176-1181. IAARC Publications. (SCOPUS)
Mingyun Kang, Sebeen Yoon, Juho Han, Sanghyeon Na, Taehoon Kim
[10.22260/ISARC2024/0152](https://doi.org/10.22260/ISARC2024/0152)  (ISARC 2024 — International Symposium on Automation and Robotics in Construction)
- Computer Vision-Based Tile Counting Model for Automated Progress Monitoring** June 2023
CCC 2023, Keszthely, Hungary.
Mingyun Kang, Wooseok Lee, Junsang Yoo, Sebeen Yoon, Taehoon Kim
(Creative Construction Conference (CCC 2023))
- A Preliminary Study on Automatic Jack Support Installation Interval Measurement Model** June 2023
CCC 2023, Keszthely, Hungary.
Sebeen Yoon, **Mingyun Kang**, Taehoon Kim
(Creative Construction Conference (CCC 2023))

Publications (Domestic)

- A Basic Study on 3D Reconstruction Performance Improvement Using 360° Camera** Nov 2024
KIBC Autumn 2024, 203-204.
Mingyun Kang, Juho Han, Sebeen Yoon, Taehoon Kim
(Proceedings of KIBC Autumn Conference)

Performance Comparison Study on Image-based 3D Space Generation AIK Fall 2024, Gyeongju, 974-975. Mingyun Kang , Sebeen Yoon, Taehoon Kim (Architectural Institute of Korea Fall Conference 2024)	Oct 2024
Applicability Enhancement of Deep Learning-based Computer Vision Models for 360° Panoramic Images KICEM 2023, Goseong, 205-206. Mingyun Kang , Junsang Yoo, Wooseok Lee, Taehoon Kim (Proceedings of KICEM Annual Conference)	Nov 2023
Computer Vision-based Adhesion Quality Inspection Automation Model for Exterior Insulation and Finishing System JKIBC, 23(2), 165-173. (KCI) Sebeen Yoon, Mingyun Kang , Hyeonseung Jang, Taehoon Kim (Journal of the Korean Institute of Building Construction (JKIBC))	Mar 2023

Patents

Automatic Supervision Method and Device for Adhesive Application Quality Domestic Patent Application No. 10-2023-0053756. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Sebeen Yoon, Mingyun Kang	Apr 2023
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Patents (Pending)

Method for Detecting Illegal Rooftop Buildings Based on Video Image Analysis Domestic Patent Application No. 10-2025-0117608. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Mingyun Kang , Seonwoo Lee, Yeon Choi, Yujin Oh, Gangmin Bae	Aug 2025
Method for Measuring Rebar Spacing Based on Single Image Information Domestic Patent Application No. 10-2024-0127210. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Mingyun Kang , Cheolhun Park, Jihyeon Lee, Jieun Lee, Yunjae Jeon	Sept 2024
Tile Quantity Counting System Domestic Patent Application No. 10-2023-0135013. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Sebeen Yoon, Mingyun Kang , Wooseok Lee, Junsang Yoo	Oct 2023
Automated Construction Site Supervision System Domestic Patent Application No. 10-2023-0125464. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation. Taehoon Kim, Sebeen Yoon, Mingyun Kang	Sept 2023

Awards

Outstanding Thesis Award Awarded for the Master's thesis: Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System Seoul National University of Science and Technology	Aug 2025
Excellence Paper Award Awarded at the Spring Conference 2025 for the paper: A Basic Study on Image Analysis-based Crackdown on Illegal Buildings on Rooftops Korean Institute of Building Construction (KIBC)	May 2025

Best Paper Award
Awarded at the Spring Conference 2024 for the paper: A Basic Study on Rebar Spacing Measurement Method
Based on Single Image Information
Korean Institute of Building Construction (KIBC)

May 2024

Best Paper Award
Awarded at the Spring Conference 2023 for the paper: A Basic Study on Image-based Numerical Measurement
Model for Construction Sites
Korean Institute of Building Construction (KIBC)

May 2023

Languages & Skills

◦ Korean (Native) · English (Advanced) · Python (Advanced)

Certificates

TOEFL iBT — Score 4.5/6.0 (90/120)
RC 5.0 / LC 5.0 / WRT 3.5 / SPK 4.5

July 2026

Architectural Engineer License

June 2025

References

Professor Taehoon Kim